1 Introduction to Categories

1.1 Motivation

Many areas of mathematics have the same basic pattern:

- Objects (such as groups, sets, vector spaces, topological spaces).
- Structure-preserving maps between those objects.

Examples:

Objects	Structure-Preserving Maps
Sets	Functions
Groups	Group Homomorphisms
Vector Spaces	Linear Transformations
Topological Spaces	Continuous Functions

Category theory studies mathematical structures through the relationships (maps) between them.

1.2 Definition of a Category

A **category** $\mathcal{C}$ consists of:

1. A collection of objects.
2. For every pair of objects A, B, a collection of morphisms (or arrows)
$$\mathrm{Hom}_{\mathcal{C}}(A, B),$$
whose elements are written
$$f : A \to B.$$
3. A composition operation
$$\mathrm{Hom}(A, B) \times \mathrm{Hom}(B, C) \longrightarrow \mathrm{Hom}(A, C),$$
given by
$$(f, g) \mapsto g \circ f.$$
4. For every object A, an identity morphism
$$1_A : A \to A.$$

These data must satisfy the following axioms.

1.3 Category Axioms

1.3.1 Associativity

For morphisms

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D,$$

we require

$$(h \circ g) \circ f = h \circ (g \circ f).$$

1.3.2 Identity

For every morphism

$$f : A \to B,$$

we have

$$f \circ 1_A = f,$$

and

$$1_B \circ f = f.$$

1.4 Example: The Category Set

The category **Set** has:

- Objects: Sets.
- Morphisms: Functions between sets.

Composition is ordinary function composition, and identity morphisms are identity functions.

Therefore **Set** is a category.

1.5 Example: The Category Grp

The category **Grp** has:

- Objects: Groups.
- Morphisms: Group homomorphisms.

For example,

$$f : \mathbb{Z} \to \mathbb{Z}_6, \qquad f(n) = n \pmod 6,$$

is a group homomorphism.

Composition is composition of homomorphisms, and identity morphisms are identity homomorphisms.

Therefore **Grp** is a category.

1.6 Example: The Category $\mathbf{Vect}_F$

The category $\mathbf{Vect}_F$ has:

- Objects: Vector spaces over a field F.
- Morphisms: Linear transformations.

Example:

$$T : \mathbb{R}^2 \to \mathbb{R}^2, \qquad T(x, y) = (x + y, y).$$

Since T is linear, it is a morphism in $\mathbf{Vect}_{\mathbb{R}}$.

1.7 Terminology

Classical Mathematics	Category Theory
Set	Object
Function	Morphism
Homomorphism	Morphism
Linear Transformation	Morphism
Continuous Function	Morphism

In category theory, the term *morphism* refers to the appropriate structure-preserving map.

1.8 Why Morphisms Matter

A central philosophy of category theory is:

> To understand an object, study how it relates to other objects through morphisms.

For example, instead of studying a group G only through its internal structure, one may study collections of morphisms

$$\mathrm{Hom}(G, H)$$

for various groups H.

1.9 Diagrams

Categories are often represented using diagrams:

$$A \xrightarrow{f} B \xrightarrow{g} C.$$

Composition produces the morphism

$$A \xrightarrow{g \circ f} C.$$

Reasoning with such diagrams is fundamental in category theory.

1.10 A Small Category

Consider a category with objects

$$\{A, B, C\}$$

and morphisms

$$1_A,\ 1_B,\ 1_C,\ f : A \to B,\ g : B \to C.$$

Closure under composition requires the morphism

$$g \circ f : A \to C.$$

This forms a category.

1.11 Key Idea

A category consists of:

- Objects,
- Morphisms between objects,

- Composition of morphisms,
- Identity morphisms,

satisfying associativity and identity laws.

The primary focus is not the objects themselves, but the network of morphisms connecting them.

2 Generators in a Category

2.1 Definition

Let $\mathcal{C}$ be a category. An object $G \in \mathcal{C}$ is called a *generator* if for every pair of distinct morphisms

$$f, g : X \to Y,$$

there exists a morphism

$$u : G \to X$$

such that

$$f \circ u \neq g \circ u.$$

Equivalently, whenever

$$f \circ u = g \circ u$$

for all morphisms $u : G \to X$, it follows that

$$f = g.$$

Thus, maps from G are sufficient to distinguish morphisms in the category.

2.2 Intuition

A generator is an object that acts as a *probe.* If two morphisms differ, then some map from the generator can detect the difference.

In many categories, morphisms from a generator correspond to the "elements" or "points" of an object.

2.3 Examples

1. The Category of Sets

In the category **Set**, any singleton set

$$\{*\}$$

is a generator.

A function

$$u : \{*\} \to X$$

is completely determined by the image of $*$, so there is a natural bijection

$$\mathrm{Hom}_{\mathbf{Set}}(\{*\}, X) \cong X.$$

If

$$f, g : X \to Y$$

are distinct, then there exists $x \in X$ such that

$$f(x) \neq g(x).$$

Define

$$u(*) = x.$$

Then

$$(f \circ u)(*) = f(x) \quad \text{and} \quad (g \circ u)(*) = g(x),$$

so

$$f \circ u \neq g \circ u.$$

Hence $\{*\}$ is a generator of **Set**.

2. The Category of Groups

In the category **Grp**, the group

$$\mathbb{Z}$$

is a generator.

If

$$f, g : G \to H$$

are distinct group homomorphisms, then there exists $a \in G$ such that

$$f(a) \neq g(a).$$

Define

$$u : \mathbb{Z} \to G, \qquad u(n) = a^n.$$

Then

$$(f \circ u)(1) = f(a) \quad \text{and} \quad (g \circ u)(1) = g(a),$$

so

$$f \circ u \neq g \circ u.$$

Therefore $\mathbb{Z}$ is a generator of **Grp**.

3. The Category of Vector Spaces

Let $\mathbf{Vect}_F$ denote the category of vector spaces over a field F.

The one-dimensional vector space

$$F$$

is a generator.

For every vector space V,

$$\mathrm{Hom}_{\mathbf{Vect}_F}(F, V) \cong V.$$

Indeed, a linear map

$$u : F \to V$$

is determined by the vector

$$u(1) \in V.$$

If

$$T, S : V \to W$$

are distinct linear maps, there exists $v \in V$ such that

$$T(v) \neq S(v).$$

Define

$$u(a) = av.$$

Then

$$(T \circ u)(1) = T(v) \quad \text{and} \quad (S \circ u)(1) = S(v),$$

so

$$T \circ u \neq S \circ u.$$

Hence F is a generator of $\mathbf{Vect}_F$.

4. The Category of R-Modules

Let R-**Mod** denote the category of left R-modules.

The module

$$R$$

(viewed as a left module over itself) is a generator.

For every R-module M,

$$\mathrm{Hom}_R(R, M) \cong M.$$

Indeed, every module homomorphism

$$u : R \to M$$

is completely determined by

$$u(1) \in M,$$

since

$$u(r) = r\,u(1).$$

If

$$f, g : M \to N$$

are distinct module homomorphisms, then there exists $m \in M$ such that

$$f(m) \neq g(m).$$

Define

$$u : R \to M, \qquad u(r) = rm.$$

Then

$$(f \circ u)(1) = f(m) \quad \text{and} \quad (g \circ u)(1) = g(m),$$

so

$$f \circ u \neq g \circ u.$$

Therefore R is a generator of R-**Mod**.

5. The Category of Topological Spaces

In the category **Top**, the one-point space

$$\{*\}$$

is a generator.

A continuous map

$$u : \{*\} \to X$$

corresponds exactly to a point of X.

Thus,

$$\mathrm{Hom}_{\mathbf{Top}}(\{*\}, X) \cong X.$$

If

$$f, g : X \to Y$$

are distinct continuous maps, there exists $x \in X$ such that

$$f(x) \neq g(x).$$

Define

$$u(*) = x.$$

Then

$$f \circ u \neq g \circ u.$$

Hence $\{*\}$ is a generator of **Top**.

2.4 Summary Table

Category	Generator	Maps from Generator Represent
Set	$\{*\}$	Elements
Grp	$\mathbb{Z}$	Group elements
$\mathbf{Vect}_F$	F	Vectors
R-**Mod**	R	Module elements
Top	$\{*\}$	Points

2.5 Key Insight

A generator allows us to study objects through morphisms from a single distinguished object. In many categories, morphisms from the generator correspond precisely to the "elements" of an object.

Thus generators convert internal structure into external morphisms, a central theme of category theory.

3 Grothendieck Categories

3.1 Definition

A **Grothendieck category** $\mathcal{A}$ is an abelian category satisfying the following conditions:

1. $\mathcal{A}$ is an **abelian category**.

2. $\mathcal{A}$ has **arbitrary (possibly infinite) coproducts**, i.e. for every family of objects $(X_i)_{i\in I}$, the coproduct

   $$\bigoplus_{i\in I} X_i$$

   exists in $\mathcal{A}$.

3. $\mathcal{A}$ satisfies **AB5**:

   filtered colimits (direct limits) of short exact sequences are exact. That is, given a directed system of short exact sequences

   $$0 \to A_i \to B_i \to C_i \to 0,$$

   the induced sequence

   $$0 \to \varinjlim A_i \to \varinjlim B_i \to \varinjlim C_i \to 0$$

   is also exact.

4. $\mathcal{A}$ has a **generator**. That is, there exists an object $G \in \mathcal{A}$ such that the functor

   $$\mathrm{Hom}(G, -) : \mathcal{A} \to \mathbf{Set}$$

   is faithful.

3.2 Generator Condition (Equivalent Form)

An object $G \in \mathcal{A}$ is a generator if and only if for every object $X \in \mathcal{A}$, there exists a set I and an epimorphism

$$G^{(I)} \twoheadrightarrow X,$$

where

$$G^{(I)} = \bigoplus_{i\in I} G$$

is the coproduct of copies of G, indexed by the set I.

3.3 Remarks

- The AB5 condition implies that filtered colimits behave well with respect to exact sequences.
- Coproducts in $\mathcal{A}$ generalize direct sums in module categories.
- The existence of a generator ensures that every object in the category can be "built" from a single object via coproducts and quotients.

4 Basic Definitions Used in Grothendieck Categories

4.1 Abelian Categories

A category $\mathcal{C}$ is called **abelian** if:

1. $\mathcal{C}$ is **preadditive**, i.e. each hom-set $\mathrm{Hom}(A, B)$ is an abelian group and composition is bilinear.
2. $\mathcal{C}$ has a **zero object**, denoted 0, which is both initial and terminal.
3. $\mathcal{C}$ has all **binary biproducts** (direct sums) $A \oplus B$.
4. $\mathcal{C}$ has all **kernels** and **cokernels**.
5. Every monomorphism and epimorphism is **normal**, i.e.
   - every monomorphism is a kernel of some morphism,
   - every epimorphism is a cokernel of some morphism.

4.2 Coproducts

Let $\mathcal{C}$ be a category and let X_1, X_2 be objects of $\mathcal{C}$.

A **coproduct** of X_1 and X_2 is an object

$$X_1 \sqcup X_2$$

together with morphisms

$$i_1 : X_1 \to X_1 \sqcup X_2, \qquad i_2 : X_2 \to X_1 \sqcup X_2,$$

satisfying the following universal property:

For every object Y and morphisms

$$f_1 : X_1 \to Y, \qquad f_2 : X_2 \to Y,$$

there exists a unique morphism

$$f : X_1 \sqcup X_2 \to Y$$

such that the diagram commutes, i.e.

$$f_1 = f \circ i_1, \qquad f_2 = f \circ i_2.$$

4.3 Universal Property Diagram

$$\begin{tikzcd}
X_1 \arrow[r, "i_1"] \arrow[dr, "f_1"'] & X_1 \sqcup X_2 \arrow[d, "f"] & X_2 \arrow[l, "i_2"'] \arrow[dl, "f_2"] \\
& Y &
\end{tikzcd}$$

4.4 Notation

The coproduct of X_1 and X_2 is often written as:

$$X_1 \sqcup X_2 \quad \text{or} \quad X_1 \oplus X_2 \quad \text{or} \quad X_1 + X_2.$$

In abelian categories, coproducts and products coincide for finite families, and are called **biproducts**.

4.5 Direct limits of short exact sequence

Let $\{0 \to A_i \to B_i \to C_i \to 0\}_{i \in I}$ be a directed system of short exact sequences in an abelian category $\mathcal{A}$, indexed by a directed set I. Taking direct limits (colimits) gives the induced sequence

$$0 \longrightarrow \varinjlim_{i \in I} A_i \longrightarrow \varinjlim_{i \in I} B_i \longrightarrow \varinjlim_{i \in I} C_i \longrightarrow 0.$$

In a Grothendieck category (AB5), this sequence is again short exact. Let I be a directed set and let

$$\left\{ 0 \to A_i \xrightarrow{f_i} B_i \xrightarrow{g_i} C_i \to 0 \right\}_{i \in I}$$

be a directed system of short exact sequences in an abelian category $\mathcal{A}$.

For each $i \le j$ in I, assume we have compatible morphisms

$$\alpha_{ij} : A_i \to A_j, \quad \beta_{ij} : B_i \to B_j, \quad \gamma_{ij} : C_i \to C_j$$

such that all structure maps commute with the short exact sequences, i.e.

$$\beta_{ij} \circ f_i = f_j \circ \alpha_{ij}, \qquad \gamma_{ij} \circ g_i = g_j \circ \beta_{ij}.$$

Step 1: Form the direct limits

Take direct limits in $\mathcal{A}$:

$$A := \varinjlim_{i \in I} A_i, \qquad B := \varinjlim_{i \in I} B_i, \qquad C := \varinjlim_{i \in I} C_i.$$

Let the canonical maps be

$$\alpha_i : A_i \to A, \qquad \beta_i : B_i \to B, \qquad \gamma_i : C_i \to C.$$

Step 2: Induced morphisms

By the universal property of colimits, the maps f_i induce a unique morphism

$$f := \varinjlim f_i : A \to B$$

such that

$$f \circ \alpha_i = \beta_i \circ f_i \quad \text{for all } i \in I.$$

Similarly, the maps g_i induce

$$g := \varinjlim g_i : B \to C$$

such that

$$g \circ \beta_i = \gamma_i \circ g_i.$$

Step 3: Resulting sequence

We obtain an induced sequence

$$A \xrightarrow{f} B \xrightarrow{g} C.$$

Step 4: Exactness after taking limits

In general abelian categories, the sequence need not be exact.

However, if $\mathcal{A}$ is a Grothendieck category (i.e. satisfies AB5), then filtered colimits are exact, and we obtain a short exact sequence:

$$0 \to \varinjlim A_i \xrightarrow{\varinjlim f_i} \varinjlim B_i \xrightarrow{\varinjlim g_i} \varinjlim C_i \to 0.$$

Summary

The process is:

direct system of SES $\implies$ take colimits objectwise $\implies$ induced sequence $\implies$ short exact seque

5 Examples of Grothendieck Categories

5.1 Abelian Groups

The category **Ab** of abelian groups is a Grothendieck category.

- It is an abelian category.
- It has arbitrary coproducts (direct sums of abelian groups).
- It satisfies AB5: filtered colimits of short exact sequences of abelian groups are exact.
- It has a generator: the group $\mathbb{Z}$.

Indeed, for any abelian group A, there exists a set I and a surjection

$$\mathbb{Z}^{(I)} \twoheadrightarrow A,$$

so $\mathbb{Z}$ is a generator.

5.2 Modules over a Ring

Let R be an associative ring with identity (not necessarily commutative). The category

$$\mathrm{Mod}(R)$$

of right (or left) R-modules is a Grothendieck category.

- It is abelian.
- It has arbitrary coproducts (direct sums of modules).
- It satisfies AB5.
- It has a generator: the free module R.

For every module M, there exists a set I and an epimorphism

$$R^{(I)} \twoheadrightarrow M.$$

5.3 Sheaves of Abelian Groups

Let X be a topological space. The category

$$\mathbf{Sh}(X)$$

of sheaves of abelian groups on X is a Grothendieck category.

More generally, for any ring R, the category of sheaves of R-modules on X is also Grothendieck.

- It is abelian.
- It has arbitrary coproducts (computed sectionwise).
- It satisfies AB5 (filtered colimits are computed stalkwise and preserve exactness).
- It has a generator (constructed from skyscraper sheaves).

5.4 Sheaves on a Ringed Space

Let $(X, \mathcal{O}_X)$ be a ringed space. Then the category

$$\mathcal{O}_X\text{-Mod}$$

of sheaves of $\mathcal{O}_X$-modules is a Grothendieck category.

- Abelian structure comes from sheaf morphisms.
- Coproducts and limits are computed sectionwise.
- AB5 holds via stalkwise exactness.
- A generator exists via direct sums of structure sheaves over open sets.

5.5 Quasi-Coherent Sheaves

Let V be an algebraic variety, scheme, or algebraic stack. The category

$$\text{Qcoh}(V)$$

of quasi-coherent sheaves is a Grothendieck category.

- Abelian structure depends on the geometry of V.
- Arbitrary coproducts exist.
- Filtered colimits are exact.
- A generator exists locally via affine charts.

5.6 Sheaves on a Site

Let $(\mathcal{C}, J)$ be a small site, i.e. a small category $\mathcal{C}$ equipped with a Grothendieck topology J.

Then the category

$$\mathbf{Sh}(\mathcal{C}, J)$$

of sheaves of abelian groups on the site is a Grothendieck category.

- It is abelian.

- It has all coproducts.
- It satisfies AB5.
- It has a generator given by representable sheaves (via Yoneda embedding and sheafification).